Project Title

Hospital Equipment Management System Using PostgreSQL

Project Description

This project focuses on designing and analyzing a hospital equipment management database using **PostgreSQL**. The database stores patient information, department details, admission records, vital signs, equipment usage, hospital stay duration, and treatment costs. SQL queries were executed through **pgAdmin 4**, the graphical administration tool for PostgreSQL, to retrieve meaningful business insights and support efficient hospital data management.

Software Used

- **Database:** PostgreSQL
- **Database Administration Tool:** pgAdmin 4
- **Query Language:** SQL

1. How many patients are admitted to the hospital?

Calculates the total number of patients admitted to the hospital.

```sql
SELECT *
FROM public.equipments;
SELECT COUNT(*) AS total_patients
FROM public.equipments;
```

Data Output Messages Notifications

	total_patients bigint
1	150

2. What is the average age of all admitted patients?

Determines the average age of all admitted patients.

```sql
SELECT *
FROM public.equipments;
SELECT AVG(age) AS average_age
FROM public.equipments;
```

Data Output Messages Notifications

	average_age numeric
1	48.4866666666666667

3. Which department has the highest number of patient admissions?

Identifies the department with the maximum patient admissions.

```sql
SELECT *
FROM public.equipments;
SELECT department, COUNT(*) AS total_patients
FROM public.equipments
GROUP BY department
ORDER BY total_patients DESC
LIMIT 1;
```

Data Output Messages Notifications

	department character varying (255)	total_patients bigint
1	General Medicine	33

4. Which department has the lowest number of patient admissions?

   Finds the department with the fewest patient admissions.

```sql
SELECT *
FROM public.equipments;
SELECT department, COUNT(*) AS total_patients
FROM public.equipments
GROUP BY department
ORDER BY total_patients ASC
LIMIT 1;
```

Data Output Messages Notifications

	department character varying (255)	total_patients bigint
1	Cardiology	18

5. How many patients are admitted to each department?

   Displays the patient count for every hospital department.

```sql
SELECT *
FROM public.equipments;
SELECT department, COUNT(*) AS total_patients
FROM public.equipments
GROUP BY department;
```

Data Output Messages Notifications

	department character varying (255)	total_patients bigint
1	ICU	25
2	Cardiology	18
3	Neurology	25
4	General Medicine	33
5	Pediatrics	24
6	Orthopedics	25

6. Which patient is the oldest?

   Retrieves the details of the oldest admitted patient.

```sql
SELECT *
FROM public.equipments
ORDER BY age DESC
LIMIT 1;
```

Data Output Messages Notifications

Showing

	patient_id character varying (100)	age integer	department character varying (255)	admission_date date	heart_rate integer
1	P1003	85	General Medicine	2026-03-01	97

7. Which patient is the youngest?

Retrieves the details of the youngest admitted patient.

```sql
SELECT *
FROM public.equipments
ORDER BY age ASC
LIMIT 1;
```

Data Output Messages Notifications

Showin

	patient_id character varying (100)	age integer	department character varying (255)	admission_date date	heart_rate integer
1	P1004	8	Orthopedics	2026-03-25	70

8. How many patients are above 60 years of age?

Counts the number of senior patients aged above 60 years.

```sql
SELECT COUNT(*) AS patients_above_60
FROM public.equipments
WHERE age > 60;
```

Data Output Messages Notifications

	patients_above_60 bigint
1	55

9. What is the average heart rate of all patients?

Calculates the average heart rate across all patients.

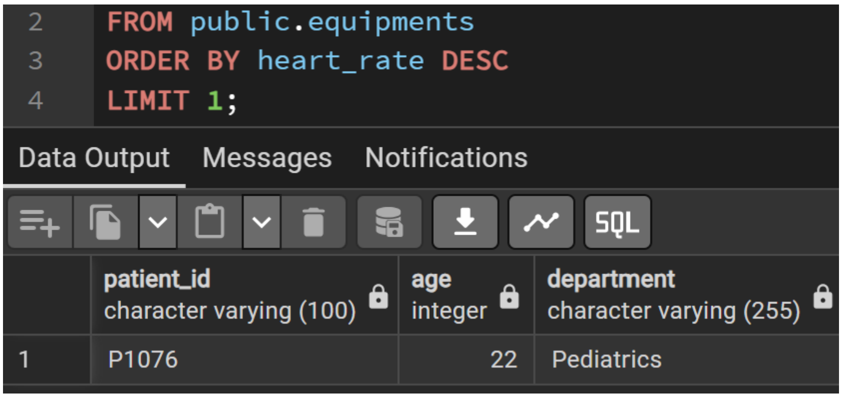
```sql
SELECT AVG(heart_rate) AS average_heart_rate
FROM public.equipments;
```

Data Output Messages Notifications

	average_heart_rate numeric
1	91.2666666666666667

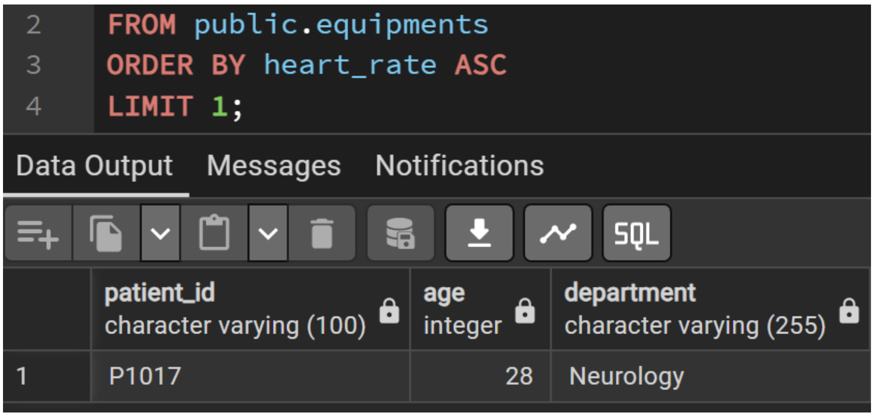
10. Which patient has the highest heart rate?

Identifies the patient with the maximum recorded heart rate.



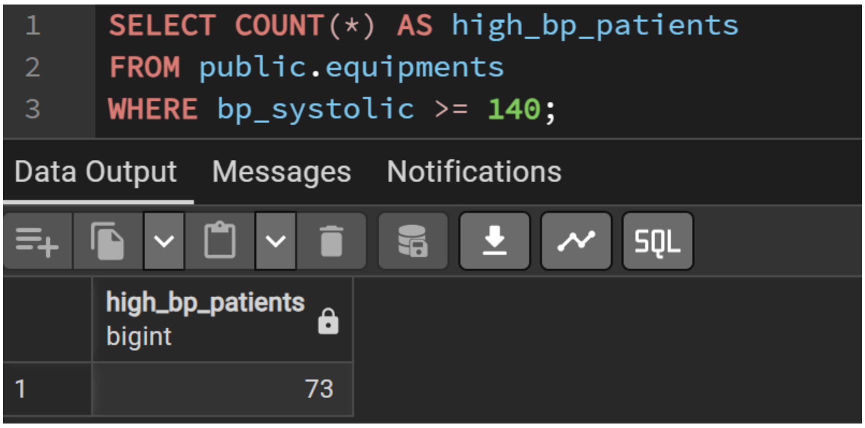
11. Which patient has the lowest heart rate?

Identifies the patient with the minimum recorded heart rate.



12. How many patients have high systolic blood pressure (≥140 mmHg)?

Counts patients diagnosed with high systolic blood pressure.



13. How many patients have low oxygen saturation (<95%)?

Counts patients with oxygen saturation below the normal level.

```sql
SELECT COUNT(*) AS low_oxygen_patients
FROM public.equipments
WHERE oxygen_saturation < 95;
```

Data Output Messages Notifications

	low_oxygen_patients bigint
1	66

14. What is the average oxygen saturation of all patients?

Computes the average oxygen saturation of admitted patients.

```sql
SELECT AVG(oxygen_saturation) AS average_oxygen
FROM public.equipments;
```

Data Output Messages Notifications

	average_oxygen numeric
1	94.3733333333333333

15. Which department has the highest average heart rate?

Determines the department with the highest average heart rate.

```sql
SELECT department,
AVG(heart_rate) AS avg_heart_rate
FROM public.equipments
GROUP BY department
ORDER BY avg_heart_rate DESC
LIMIT 1;
```

Data Output Messages Notifications

	department character varying (255)	avg_heart_rate numeric
1	Pediatrics	105.0416666666666667

16. Which department has the highest average treatment cost?

Identifies the department with the highest average treatment expenses.

```sql
SELECT department,
AVG(treatment_cost_inr) AS avg_cost
FROM public.equipments
GROUP BY department
ORDER BY avg_cost DESC
LIMIT 1;
```

Data Output Messages Notifications

	department character varying (255)	avg_cost numeric
1	ICU	96887.000000000000

17. What is the average treatment cost for each department?

Calculates the average treatment cost department-wise.

```sql
SELECT department,
AVG(treatment_cost_inr) AS average_cost
FROM public.equipments
GROUP BY department;
```

Data Output Messages Notifications

	department character varying (255)	average_cost numeric
1	ICU	96887.000000000000
2	Cardiology	37020.777777777778
3	Neurology	41859.280000000000
4	General Medicine	16520.121212121212
5	Pediatrics	16582.583333333333
6	Orthopedics	27192.240000000000

18. Which patient has the highest treatment cost?

Finds the patient with the highest treatment cost.

```sql
SELECT *
FROM public.equipments
ORDER BY treatment_cost_inr DESC
LIMIT 1;
```

Data Output Messages Notifications

	patient_id character varying (100)	age integer	department character varying (255)
1	P1090	69	ICU

19. Which patient has the lowest treatment cost?

Finds the patient with the lowest treatment cost.

```sql
SELECT *
FROM public.equipments
ORDER BY treatment_cost_inr ASC
LIMIT 1;
```

Data Output Messages Notifications

	patient_id character varying (100)	age integer	department character varying (255)
1	P1059	50	General Medicine

20. Which medical equipment is used most frequently?

Identifies the most frequently used medical equipment.

```sql
SELECT equipment_used,
COUNT(*) AS usage_count
FROM public.equipments
GROUP BY equipment_used
ORDER BY usage_count DESC
LIMIT 1;
```

Data Output Messages Notifications

	equipment_used character varying (255)	usage_count bigint
1	Infusion Pump	32

21. How many patients use each type of medical equipment?

Counts the number of patients using each type of equipment.

```sql
SELECT equipment_used,
COUNT(*) AS total_patients
FROM public.equipments
GROUP BY equipment_used;
```

Data Output Messages Notifications

	equipment_used character varying (255)	total_patients bigint
1	Defibrillator	20
2	BP Monitor	25
3	ECG Monitor	26
4	Ventilator	25
5	Pulse Oximeter	22
6	Infusion Pump	32

22. Which department uses the most medical equipment?

Determines the department with the highest equipment utilization.

```sql
SELECT department,
COUNT(equipment_used) AS equipment_count
FROM public.equipments
GROUP BY department
ORDER BY equipment_count DESC
LIMIT 1;
```

Data Output Messages Notifications

	department character varying (255)	equipment_count bigint
1	General Medicine	33

23. What is the average length of hospital stay?

Calculates the average duration of patient hospitalization.

```sql
SELECT AVG(length_of_stay_days) AS average_stay
FROM public.equipments;
```

Data Output Messages Notifications

	average_stay numeric
1	5.1333333333333333

24. Which patient has the longest hospital stay?

Identifies the patient with the longest hospital stay.

```sql
SELECT *
FROM public.equipments
ORDER BY length_of_stay_days DESC
LIMIT 1;
```

Data Output Messages Notifications

	patient_id character varying (100)	age integer	department character varying (255)
1	P1116	65	ICU

25. Which department has the longest average patient stay?

Finds the department where patients stay the longest on average.

```sql
SELECT department,
AVG(length_of_stay_days) AS avg_stay
FROM public.equipments
GROUP BY department
ORDER BY avg_stay DESC
LIMIT 1;
```

Data Output Messages Notifications

	department character varying (255)	avg_stay numeric
1	ICU	9.0000000000000000

26. How many patients were admitted on March 15?

Counts patients admitted after the specified admission date.

```sql
SELECT COUNT(*) AS total_patients
FROM public.equipments
WHERE admission_date > '2026-03-15';
```

Data Output Messages Notifications

	total_patients bigint
1	80

27. Which patients have abnormal vital signs (high heart rate, high blood pressure, or low oxygen saturation)?

Identifies patients with abnormal heart rate, blood pressure, or oxygen saturation.

```sql
SELECT patient_id,
heart_rate,
bp_systolic,
oxygen_saturation
FROM public.equipments
WHERE heart_rate > 100
OR bp_systolic >= 140
OR oxygen_saturation < 95;
```

Data Output Messages Notifications

	patient_id character varying (100)	heart_rate integer	bp_systolic integer	oxygen_saturation integer
1	P1001	127	126	95
2	P1002	117	99	96
3	P1003	97	164	95
4	P1004	70	149	99
5	P1005	84	140	94

28. Which patients have treatment costs higher than the hospital's average treatment cost?

Lists patients whose treatment costs exceed the hospital average.

```sql
SELECT patient_id,
treatment_cost_inr
FROM public.equipments
WHERE treatment_cost_inr >
(
    SELECT AVG(treatment_cost_inr)
    FROM public.equipments
);
```

Data Output Messages Notifications

	patient_id character varying (100)	treatment_cost_inr integer
1	P1007	53238
2	P1010	68352
3	P1011	114341
4	P1014	122240
5	P1017	51271

29. What are the top five patients with the highest treatment costs?

Displays the top five patients with the highest treatment expenses.

```sql
SELECT patient_id,
treatment_cost_inr
FROM public.equipments
ORDER BY treatment_cost_inr DESC
LIMIT 5;
```

Data Output Messages Notifications

	patient_id character varying (100)	treatment_cost_inr integer
1	P1090	149134
2	P1020	146957
3	P1046	138235
4	P1053	134297
5	P1071	126900